

Sleeping Tumors

The Biology of Cancer Dormancy and Why Disseminated Cells Relapse Years Later

Omar Nossair — Literature Review

Introduction

Cancer recurrence years or even decades after apparently successful treatment remains one of oncology's most difficult open problems. Surgeons have long observed what is sometimes called the Halsted paradox: removing a primary tumor completely, with clean margins and no detectable spread, does not guarantee cure. A meaningful fraction of patients relapse with metastatic disease years after their last clean scan, in organs their original tumor never visibly touched. Clinical cohorts show that estrogen-receptor-positive (ER+) breast cancer patients frequently develop bone metastasis eight to ten years after diagnosis, with as many as 20% of early-stage patients eventually experiencing distant metastasis after this extended latency. Prostate cancer shows an even more extreme version of the same pattern, with bone recurrence documented decades after curative-intent surgery or radiation in some patients. Standard exponential growth models of cancer cannot account for a multi-year, sometimes multi-decade, gap between apparent cure and detectable relapse; something has to be holding those cells in check during the interval.

A growing body of evidence points to tumor dormancy, a reversible, non-proliferative survival state that disseminated cancer cells can enter and persist in for extended periods, as the central driver of this delayed relapse. This review works through the two recognized forms of dormancy, the molecular signaling that locks a disseminated cell into quiescence, the bone marrow niche that has become the best-studied model system for the phenomenon, how the timeline and mechanisms of dormancy differ across cancer types, what is known about how dormant cells reawaken, and where detection and treatment strategies currently stand.

1. Two Forms of Dormancy

The literature generally separates dormancy into two categories that can act together or independently. Tumor-mass dormancy describes a microscopic cluster of cells that is proliferating and dying at roughly equal rates, so the lesion's net size stays flat. This category

splits further into two sub-mechanisms. Angiogenic dormancy occurs when a micrometastasis cannot recruit its own blood supply and is capped in size by the diffusion limits of oxygen and nutrients, the mechanism first described by Judah Folkman's angiogenesis research; the lesion is not asleep so much as starved into a stalemate. Immune-mediated dormancy occurs when the immune system actively holds a micrometastasis in check, killing cells roughly as fast as they divide, often framed within an elimination–equilibrium–escape model in which dormancy corresponds to a prolonged equilibrium phase before any eventual immune escape.

Cellular dormancy, by contrast, describes a single disseminated cell, or a small number of cells, entering a reversible G0/G1 cell-cycle arrest and simply ceasing to divide, independent of blood supply or active immune killing. This is the form most closely tied to the multi-year, and in prostate cancer sometimes multi-decade, latency seen clinically, and the one with the most developed molecular map, discussed next.

2. The Molecular Switch Behind Cellular Dormancy

Work using breast and prostate cancer models has identified a fairly specific signaling switch that tips a disseminated cell into or out of cellular dormancy. A high ratio of p38 (a stress-activated kinase) to ERK (a proliferation-driving kinase) signaling is strongly associated with quiescence, while a lower p38:ERK ratio favors renewed proliferation; this ratio functions less like an on/off switch and more like a rheostat that niche signals can push in either direction (Fluegen et al., 2017; Sosa et al., 2015).

Downstream of this switch, the orphan nuclear receptor NR2F1 drives global chromatin repression, working together with the transcription factors SOX9 and RAR-beta to lock in a quiescence program. NR2F1 also induces the pluripotency gene NANOG, and experimentally blocking NR2F1 is sufficient to interrupt dormancy and allow disseminated cells to resume growth across multiple organ sites (Sosa et al., 2015). Low-oxygen conditions in the primary tumor can additionally pre-condition disseminated cells toward dormancy by inducing NR2F1 alongside hypoxia-inducible factor 1-alpha (HIF1 α); acting together, the two factors raise levels of the cell-cycle inhibitor p27 while suppressing the cell-cycle driver CDK4, sustaining arrest, an effect more pronounced in estrogen-receptor-positive breast cancer cells than in triple-negative subtypes.

Dormancy is not purely a tumor-cell-intrinsic decision; it is imposed, and actively maintained, by signals from the surrounding niche. In bone marrow, a specific population of NG2-positive/Nestin-positive mesenchymal stem cells secretes TGF-beta2 and BMP7, which

signal through TGF-beta receptor III and BMP receptor II on disseminated cells to activate the same p38/p27 axis and enforce quiescence. Depleting these niche cells experimentally is sufficient to trigger metastatic outgrowth, even with no other change to the tumor cells themselves (Bragado et al., 2021). Prostate cancer appears to use an overlapping but distinct version of this same circuit: differentiated osteoblasts secrete TGF-beta2 and GDF10, which activate TGF-beta receptor III signaling to trigger p38MAPK phosphorylation; nuclear p-p38 then phosphorylates the retinoblastoma protein at a specific site (pS249/T252), raising p27 levels and blocking G1-to-S cell-cycle progression (Yu-Lee et al., 2018). Osteoblast-secreted Wnt5a and BMP7 have separately been shown to raise p21 expression and inhibit prostate cancer cell growth within bone, reinforcing dormancy through a partially independent route (Kobayashi et al., 2011; Yumoto et al., 2016).

3. The Bone Marrow Niche as a Model System

Bone marrow has become the best-characterized dormancy niche because breast and prostate cancer cells disseminate there early and often, and because the marrow's own biology, built to keep hematopoietic stem cells quiescent until needed, turns out to be reused by cancer cells almost wholesale. Near bone-forming osteoblasts, suppressive signals in the endosteal niche help maintain high NR2F1 expression and a non-proliferative, non-glycolytic metabolic program in nearby disseminated cells. Near blood vessels, a different balance of signals applies in the perivascular niche; disseminated cells here can more easily transition toward outgrowth once local angiogenic activity picks up (Price et al., 2016).

Age-related changes in this niche have been proposed as a mechanistic link to rising late-relapse risk. As osteoblast populations decline with age, the marrow's quiescence-maintaining signals weaken, destabilizing p38 signaling and lowering NR2F1 expression. This removes the epigenetic brake on proliferation at roughly the same time that the local metabolic environment shifts to favor tumor cell growth. Bone marrow adipocytes, which expand with age and with osteoporosis, have separately been implicated as a metabolic support system for reactivating cells, supplying lipids and adipokines that dormant cells appear able to exploit once quiescence signaling weakens.

4. Dormancy Across Cancer Types

The duration and character of dormancy varies considerably by tumor type, and this variation is itself informative about the underlying biology. Lung cancer bone metastases typically show a

markedly shortened dormancy window of roughly one to three years, with most recurrences occurring within a few years of diagnosis. Breast cancer, particularly the ER-positive subtype, more often shows an extended dormant phase, with recurrence commonly manifesting eight to ten years after diagnosis and occasionally much later. Prostate cancer is the most extreme example documented in the literature: disseminated prostate cancer cells can remain dormant in bone marrow for months to decades, with the capacity for very late recurrence considered relatively unique to prostate cancer alongside ER-positive breast cancer. Because disseminated tumor cells have been detected in bone marrow even in patients with localized, apparently curable disease, dormancy is now understood as a step that can begin well before a cancer is ever classified as metastatic.

5. What Triggers Reawakening

Several distinct signals have been implicated in reactivating dormant cells, spanning vascular, skeletal, hormonal, immune, and treatment-related routes. Renewed vessel growth near a dormant lesion releases periostin alongside VEGF and placental growth factor (PlGF), tipping cells out of angiogenic dormancy and into active outgrowth. Within the endosteal niche, factors including TGF-beta1, bone morphogenetic proteins, and interleukin-1 beta are thought to promote micrometastatic outgrowth; direct tumor cell–osteoblast interactions can also increase E-cadherin expression and mTOR activity in the cancer cells, further favoring growth. Bone itself is not a static scaffold: osteoclast-mediated bone resorption liberates growth factors such as TGF-beta and IGF-1 that had been sequestered in the bone matrix, and this liberated growth-factor pool is increasingly recognized as a direct fuel source for dormancy escape.

Hormonal signaling plays a well-documented role in estrogen-receptor-positive disease: estradiol and progesterone can act directly on dormant malignant cells to promote arousal. Counterintuitively, estrogen-deprivation therapy itself can trigger reawakening in some models. Withdrawing estrogen causes the bone marrow niche to overexpress angiopoietin-2, which reactivates dormant ER-positive cells partly through the endothelial Tie2 receptor and partly through a tumor-cell integrin (Ren et al., 2018). Immune surveillance itself can flip from restraining dormancy to enabling escape: immunosuppressive myeloid-derived suppressor cells, regulatory T cells, and inflammatory triggers such as neutrophil extracellular traps have all been implicated in tipping previously immune-restrained dormant cells toward outgrowth. Certain chemotherapy regimens separately provoke release of pro-tumor cytokines such as interleukin-6 and G-CSF, which can stimulate outgrowth of previously dormant cells elsewhere in the body, a possible contributor to post-treatment relapse patterns. Finally, a changed extracellular matrix

composition at the distant site can itself trigger renewed proliferation via ECM-receptor interactions, converging on the same growth pathways, PI3K/Akt/mTOR, MAPK/ERK, Wnt/beta-catenin, and Notch signaling, that drive proliferation more broadly in cancer.

6. Detecting Dormant Disease

Disseminated tumor cells can be identified directly in bone marrow aspirates by immunofluorescence, and NR2F1 expression level has been proposed as a way to stratify which detected DTCs are more deeply dormant versus closer to reawakening, though this remains a research tool rather than a standard clinical test (Borgen et al., 2018). Blood-based circulating tumor DNA (ctDNA) assays are increasingly used to detect minimal residual disease after curative-intent treatment across cancer types, including breast, colorectal, lung, and prostate cancer. Multiple active trials, including tumor-informed panels in early breast cancer and post-prostatectomy ctDNA monitoring, are testing whether ctDNA-guided surveillance can catch recurrence earlier than imaging alone, or safely allow some patients to avoid further treatment (Liu et al., 2025). Sensitivity, assay standardization across labs, and the biological question of whether a dormant single cell sheds detectable DNA at all remain open problems limiting how far these tools have moved into routine practice.

7. Therapeutic Strategies Under Investigation

Two broadly opposite strategies are being explored, alongside efforts to make existing surveillance tools clinically actionable. A “lock and maintain” approach aims to reinforce the same signals that already keep dormant cells quiescent, for instance by therapeutically sustaining p38 activity or NR2F1 expression, turning dormancy into a durable, indefinite state rather than a temporary reprieve. The opposite “wake and kill” strategy deliberately forces dormant cells to re-enter the cell cycle under controlled conditions, since actively dividing cells are far more vulnerable to conventional chemotherapy than quiescent ones; the challenge is doing this safely without simply seeding new, now actively growing, metastases.

A third direction targets the niche rather than the tumor cell itself: because niche-derived cytokines such as TGF-beta2 and BMP7, and niche cell populations such as NG2-positive/Nestin-positive mesenchymal stem cells, appear necessary to maintain dormancy, some proposed strategies focus on protecting or restoring the niche rather than acting on the cancer cells directly. A related line of work targets the immune side of the equation, aiming to reinforce the immunosurveillance mechanisms that hold micrometastases

in equilibrium before they can escape into active growth. Separately, ongoing ctDNA-guided surveillance trials in colorectal, prostate, and other cancers are testing whether ctDNA-negative patients can safely undergo a “watch and wait” approach instead of immediate additional therapy, which would let MRD status directly guide treatment intensity rather than treating everyone the same way after primary treatment.

Limitations and Open Questions

Despite the detailed mechanistic map built in breast and prostate cancer bone marrow models, several gaps remain before this biology reliably changes patient care. It is not yet clear whether the same signaling logic generalizes across cancer types and metastatic sites beyond bone, whether any dormancy biomarker is sensitive and specific enough for routine clinical use, or whether reactivation can be safely controlled in either direction without triggering the outgrowth clinicians are trying to prevent. Much of the mechanistic evidence summarized here comes from preclinical and largely single-cancer-type models; prospective clinical validation across a broader range of tumor types remains an open and active area of research, as does the question of how to translate a growing mechanistic understanding of dormancy escape into an interventable clinical target.

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